Matthew Jovero

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Seeking a design engineering internship position starting in 2027.

Education

SAN JOSÉ STATE UNIVERSITY, San José, CA — *Expected graduation: May 2028*

***B.S. Aerospace Engineering**, Junior* — *GPA: 3.5/4.0*

Projects

SJSU 2025 AIAA DESIGN/BUILD/FLY | *Aircraft Design Engineering* — *Sept 2024 – May 2025*

- Built a MATLAB analysis that established a 25 lb. aircraft weight, 15 lb. payload, and a 70-85 Wh battery design.
- Designed a 30-inch Clark-Y wing in Inventor with beam spars and lightening holes to correct weight and strength.
- Performed FEA in Ansys to validate internal wing structure when predicting distributed rib loads from 2-8 N.
- Developed 6 GD&T drawings for DFM/DFA structural components to support aircraft manufacturing/assembly.
- Implemented sweeps and interpolation for trade-study graphs evaluating ideal aircraft speeds and energy usage.

ROBOTICS COMPETITION – 1ST PLACE | *Mechanical Design Engineering* — *Mar 2025 – May 2025*

- Managed a team of 8 to build a robot that autonomously transports beacons in 19 seconds (university record).
- Increased throughput by 52% by redesigning the chassis to improve the layout of sensors and control systems.
- Designed a motor-driven mechanical arm using C-channels to maintain strength while satisfying a weight limit.
- Programmed RobotC autonomous behaviors such as beacon tracking using sensor data feedback.

WIND TURBINE DESIGN | *Structural Design Engineering* — *Jan 2025 – Mar 2025*

- Led the design of a NACA 4424 wind turbine in SolidWorks to achieve 2 W of power under 15 N/mm stiffness.
- Performed mesh convergence FEA in Ansys to predict deformation within 5% of measured results.
- Operated drill presses, miter saws, and hand tools to prevent wood splitting and ensure part quality.

ATX CASE DESIGN | *Product Design Engineering* — *May 2026 – July 2026*

- Designed a computer tower for ATX motherboards consisting of 40+ parts in SolidWorks with DFM/DFA utility.
- Satisfied functional design factors such as a 120-140 mm fan size, 240 mm radiator, and 25 mm of cable space.

HIGH-POWER ROCKET DESIGN | *Rocketry Design Engineering – Rocket Club* — *Oct 2025 – Dec 2025*

- Designed a high-power rocket in SolidWorks, reducing cost by 12% when compared to a commercial rocket kit.
- Evaluated FEA stress and deformation from landing impact loads to iterate the most critical ground-strike fin.
- Performed tolerance stack-up analysis to solve assembly issues for the upper body components.
- Developed an exploded view diagram and 10 detailed part drawings using GD&T and BOM.

SHEET METAL WORKSHOP | *Sheet Metal Designer* — *Jan 2026 – June 2026*

- Designed 36 sheet metal components and drawings in Fusion 360 for storage and workshop applications.
- Validated sheet metal designs in Ansys by evaluating structural performance to guide design iterations.

Work Experience

MATHEMATICS TUTOR | *Hybrid STEM Teacher* — *Oct 2024 – May 2026*

- Delivered 500+ hours of instruction to guide 10+ students through solution steps for engineering mathematics.
- Built 15+ work assignments on LaTeX to improve student exam performance and problem-solving strategies.
- Provided online tutoring for international students, adapting to different notations and terminologies.

AHS MACHINE WOODSHOP | *Student Shop Assistant* — *Nov 2023 – May 2024*

- Trained new students on using different machines and equipment safely for their technical projects.
- Maintained condition of the shop and its machines (bandsaws, table saws, drill presses, toolbox, 3D printers).

Skills

Software: SolidWorks, Autodesk Inventor, Fusion 360, Ansys, FEA, MATLAB, RobotC, LaTeX, MS Office Suite

Hardware: Mechanical assembly, machine tools, epoxy bonding, sensor integration, woodwork, 3D printing (FDM)